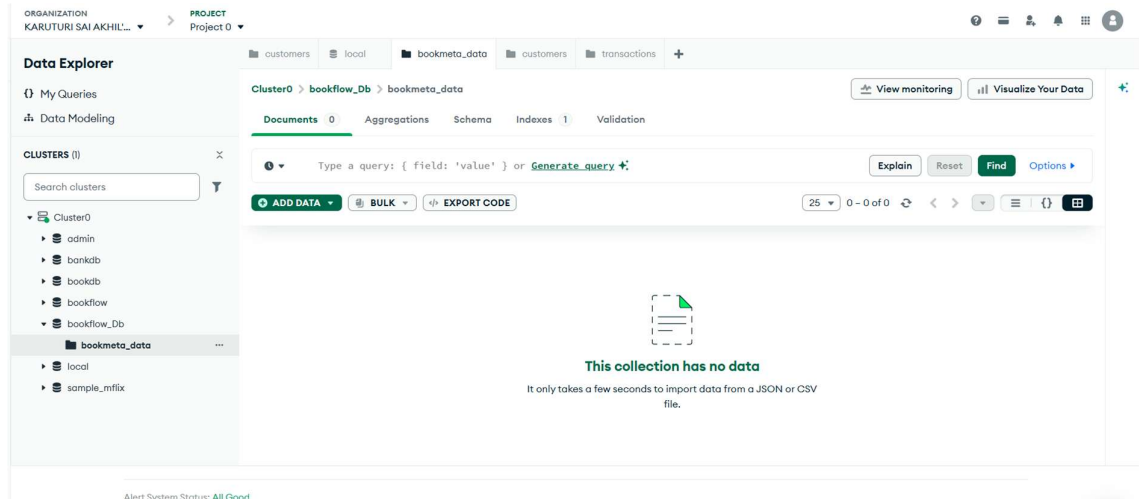


WEEK 3

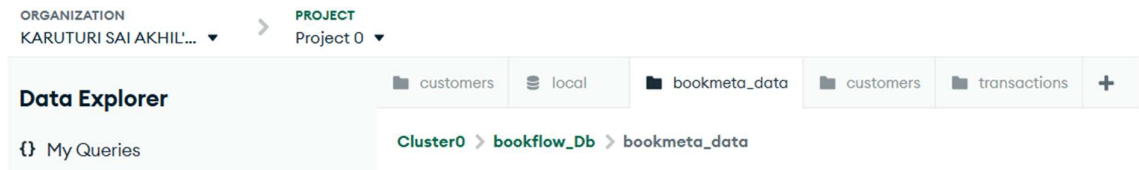
K.SAI AKHIL

2520030021

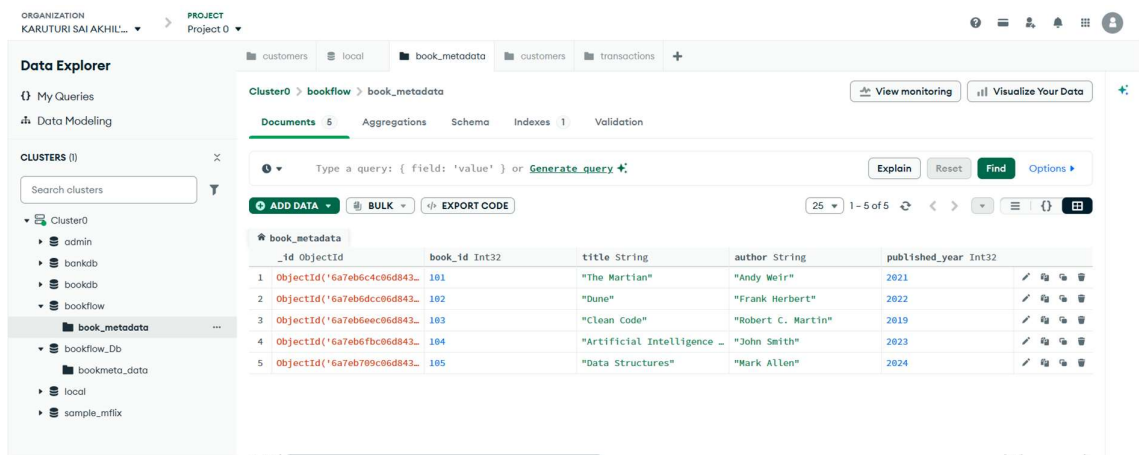
1) Create a database bookflow



2) Create a new collection book_metadata



3) Click add data and insert documents



4) Mongo DB Queries:

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Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bankab
- bookab
- bookflow
- book_metadata**
- bookflow_Db
- bookmeta_data
- local
- sample_mflix

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

Generate query Explain Reset Find Options

```
{
  "reviews.rating": {
    "$gt": 4
  }
}
```

ADD DATA BULK EXPORT CODE

25 1-5 of 5

_id ObjectId	book_id Int32	title String	author String	published_year Int32
ObjectId('6a7eb6c4c06d843...')	101	"The Martian"	"Andy Weir"	2021
ObjectId('6a7eb6d8cc06d843...')	102	"Dune"	"Frank Herbert"	2022
ObjectId('6a7eb6e0cc06d843...')	103	"Clean Code"	"Robert C. Martin"	2019
ObjectId('6a7eb6fbc06d843...')	104	"Artificial Intelligence ..."	"John Smith"	2023
ObjectId('6a7eb709c06d843...')	105	"Data Structures"	"Mark Allen"	2024

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Data Explorer

My Queries

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- bookflow_Db
- bookmeta_data
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- sample_mflix

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

Generate query Explain Reset Find Options

```
{
  "reviews.rating": {
    "$in": [4, 5]
  }
}
```

ADD DATA BULK EXPORT CODE

25 1-5 of 5

_id ObjectId	book_id Int32	title String	author String	published_year Int32
ObjectId('6a7eb6c4c06d843...')	101	"The Martian"	"Andy Weir"	2021
ObjectId('6a7eb6d8cc06d843...')	102	"Dune"	"Frank Herbert"	2022
ObjectId('6a7eb6e0cc06d843...')	103	"Clean Code"	"Robert C. Martin"	2019
ObjectId('6a7eb6fbc06d843...')	104	"Artificial Intelligence ..."	"John Smith"	2023
ObjectId('6a7eb709c06d843...')	105	"Data Structures"	"Mark Allen"	2024

5)Aggregations

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Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

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- bookab
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- book_metadata**
- bookflow_Db
- bookmeta_data
- local
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Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

View monitoring Visualize Your Data

Smatch

Generate aggregation Explain Run Options

Untitled - modified SAVE CREATE NEW EXPORT CODE

PREVIEW STAGES TEXT WIZARD

Stage1 \$match

```
1 {
2   published_year: {
3     $gt: 2020
4   }
5 }
```

Output preview after \$match stage (Sample of 4 documents)

```
_id: ObjectId('6a7eb6c4c06d843befdd3fd')
book_id: 101
title: "The Martian"
author: "Andy Weir"
published_year: 2021
category: "Science Fiction"
format: "Digital PDF"
file_size: "2.5 MB"
member_id: 101
```

```
_id: ObjectId('6a7eb6d8cc06d843b...')
book_id: 102
title: "Dune"
author: "Frank Herbert"
published_year: 2022
category: "Science Fiction"
format: "Digital EPUB"
file_size: "3.1 MB"
member_id: 102
```

ORGANIZATION KARUTURI SAI AKHIL... PROJECT Project 0

Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - admin
 - bankdb
 - bookdb
 - bookflow
 - book_metadata
 - bookflow_Db
 - bookmeta_data
 - local
 - sample_mflix

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

Smatch Sunwind

Generate aggregation + Explain Run Options

Untitled - modified SAVE + CREATE NEW EXPORT CODE

PREVIEW STAGES TEXT WIZARD

1 "\$reviews{"
2 }

Output preview after Sunwind stage (Sample of 10 documents)

```
{
  "_id": "ObjectId('6a7eb6c4c06d843befdd3fd3f')",
  "book_id": 101,
  "title": "The Martian",
  "author": "Andy Weir",
  "published_year": 2021,
  "category": "Science Fiction",
  "format": "Digital PDF",
  "file_size": "2.5 MB",
  "member_id": 101
}
```

+ Add stage

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

Smatch Sunwind \$group

Generate aggregation + Explain Run Options

Untitled - modified SAVE + CREATE NEW EXPORT CODE

PREVIEW STAGES TEXT WIZARD

Stage 3 \$group

```
1 {
2   "_id": "$title",
3   "average_rating": {
4     "$avg": "$reviews.rating"
5   },
6   "total_reviews": {
7     "$sum": 1
8   }
9 }
```

Output preview after \$group stage (Sample of 4 documents)

```
{
  "_id": "Artificial Intelligence Basics",
  "average_rating": 4.333333333333333,
  "total_reviews": 3
}
```

```
{
  "_id": "The Martian",
  "average_rating": 4.666666666666667,
  "total_reviews": 3
}
```

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

Smatch Sunwind \$group \$sort

Generate aggregation + Explain Run Options

Untitled - modified SAVE + CREATE NEW EXPORT CODE

PREVIEW STAGES TEXT WIZARD

Stage 4 \$sort

```
1 {
2   "average_rating": -1
3 }
```

Output preview after \$sort stage (Sample of 4 documents)

```
{
  "_id": "The Martian",
  "average_rating": 4.666666666666667,
  "total_reviews": 3
}
```

```
{
  "_id": "Dune",
  "average_rating": 4.666666666666667,
  "total_reviews": 3
}
```

5)Indexes

ORGANIZATION KARUTURI SAI AKHIL... PROJECT Project 0

Data Explorer

My Queries
Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - admin
 - bankdb
 - bookdb
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 - bookmeta_data
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 - sample_mflix

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema **Indexes 2** Validation

Create Index Refresh

Looking for search indexes?
These indexes can be created and viewed under [Search and Vector Search](#).

Name & Definition	Type	Size	Usage	Properties	Status
id	REGULAR	36.9 kB	1 (since Mon Aug 24 2026)	UNIQUE	READY
reviews.comments_text	TEXT	20.5 kB	0 (since Wed Aug 26 2026)	HIDDEN	READY
_fts (text) _ftsx					

6) Search using text

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Data Explorer

My Queries
Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
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 - bookdb
 - bookflow
 - book_metadata**
 - bookflow_Db
 - bookmeta_data
 - local
 - sample_mflix

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 2 Validation

View monitoring Visualize Your Data

Generate query + Explain Reset Find Options

```
{
  "$text": {
    "$search": "excellent"
  }
}
```

ADD DATA BULK EXPORT CODE

25 1-5 of 5

book_metadata	_id	ObjectId	book_id	Int32	title	String	author	String	published_year	Int32
1	ObjectId('6a7eb6c4c06d843...')	101	"The Martian"	"Andy Weir"	2021					
2	ObjectId('6a7eb6e4cc06d843...')	103	"Clean Code"	"Robert C. Martin"	2019					
3	ObjectId('6a7eb6dccc06d843...')	102	"Dune"	"Frank Herbert"	2022					
4	ObjectId('6a7eb709c06d843...')	105	"Data Structures"	"Mark Allen"	2024					
5	ObjectId('6a7eb6fbc06d843...')	104	"Artificial Intelligence ..."	"John Smith"	2023					

8) Search for a specific word

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Data Explorer

My Queries
Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - admin
 - bankdb
 - bookdb
 - bookflow
 - book_metadata**
 - bookflow_Db
 - bookmeta_data
 - local
 - sample_mflix

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 2 Validation

View monitoring Visualize Your Data

Generate query + Explain Reset Find Options

```
{
  "$text": {
    "$search": "\"science fiction\""
  }
}
```

ADD DATA BULK EXPORT CODE

25 1-2 of 2

book_metadata	_id	ObjectId	book_id	Int32	title	String	author	String	published_year	Int32
1	ObjectId('6a7eb6dccc06d843...')	102	"Dune"	"Frank Herbert"	2022					
2	ObjectId('6a7eb6c4c06d843...')	101	"The Martian"	"Andy Weir"	2021					